Ranking-Based Learning for Robust Tactile Roughness Perception under Domain Shift

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***Abstract*— Tactile roughness perception is commonly formulated as either classification or regression, implicitly assuming that tactile signals can be grounded to a stable absolute scale. In practice, however, tactile measurements are highly sensitive to sensing variability, making this numerical grounding unreliable under limited supervision or domain shift.**

In this paper, we investigate how the choice of learning objective affects robustness when absolute grounding is unstable. We compare classification, regression, and ranking-based learning on a sandpaper dataset with 5 grit levels and 9,595 samples. Under well-conditioned in-domain settings, all paradigms achieve comparable performance. However, when supervision is scarce (e.g., 5% training data) or when substantial domain shift is introduced through changes in sensor hardware and exploration protocol, their behavior diverges.

Under domain shift, regression exhibits a pronounced collapse in ordinal consistency, whereas ranking-based learning preserves relative ordering more effectively, outperforming regression by 0.290 in Spearman rank correlation. These findings suggest that directly optimizing ordinal consistency yields more robust generalization when tactile roughness labels provide reliable ordering but unstable numerical scales.

I. Introduction

Tactile sensing plays a central role in robotic manipulation by enabling robots to infer surface and material properties that are difficult or impossible to recover visually, such as texture, friction, and compliance. Recent advances in high-resolution vision-based tactile sensors—including GelSight [1], GelSlim [2], DIGIT [3], TacTip [4], and SoftBubble [5]—have significantly expanded the capability of learning-based tactile perception. These sensors provide dense, image-like contact information, enabling data-driven models to predict material categories and physical properties with increasing accuracy.

Despite these advances, tactile measurements remain inherently sensitive to sensing conditions. Many modern tactile sensors rely on soft, deformable contact layers to transduce physical contact into measurable signals. As a result, even sensor units built to the same nominal design can exhibit non-negligible response differences due to manufacturing tolerances, fabrication inconsistencies, and assembly variations. Such unit-to-unit variability arises not only in hand-crafted prototypes but also in industrial production. In addition,

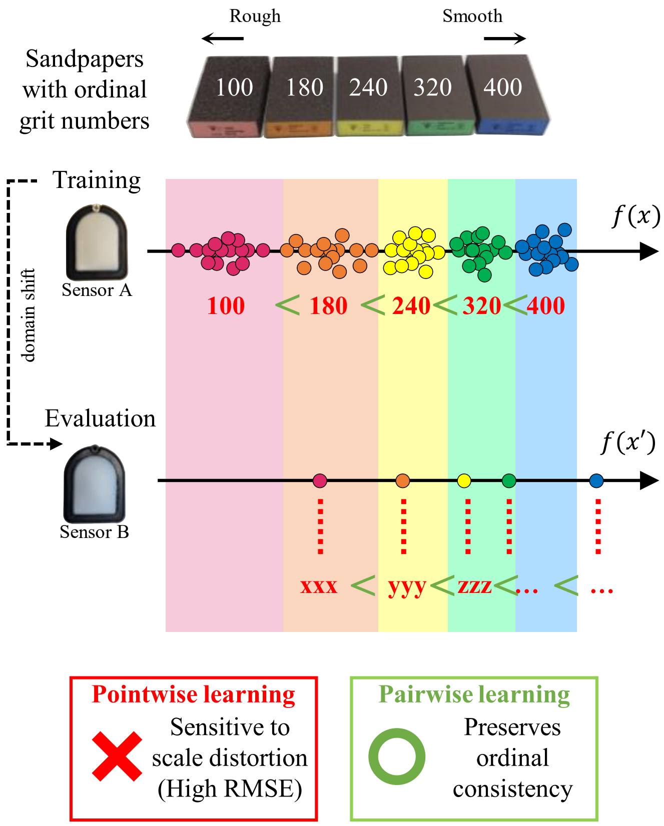

Fig. 1. **Conceptual illustration of the proposed motivation.** A model trained on Sensor A may produce shifted or distorted absolute scores when evaluated on Sensor B. Pointwise learning is sensitive to this scale distortion, whereas pairwise ranking focuses on preserving the ordinal relation among sandpaper grit levels.

variations in applied force, contact angle, and exploration strategy can substantially alter tactile responses. Together, these factors introduce practical domain shifts that challenge the stability of learned tactile representations [6], [7]. Recent reviews further emphasize that domain shift and sensor-specific calibration remain central obstacles to deploying tactile perception systems in real-world settings [8], [9].

Crucially, this variability undermines the assumption that tactile signals can be grounded to a stable absolute numerical scale. Psychophysical studies of human haptic perception reveal a related phenomenon: perceived roughness is strongly context-dependent and influenced by contact force and exploration dynamics [10]–[12]. While absolute magnitude judgments vary across conditions and individuals, relative comparisons between surfaces tend to be more consistent [13], [14]. These findings suggest that under contextual

*This work was supported in part by the National Research Council of Science and Technology as part of the project titled “Development of core technologies for robot general purpose task artificial intelligence (RoGeTA) framework” under Grant NK261B.

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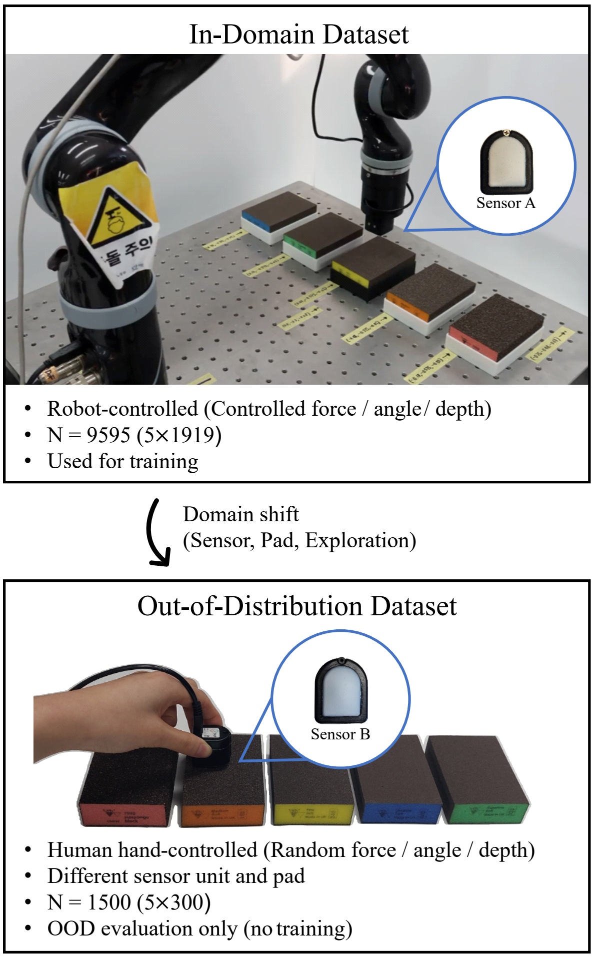

Fig. 2. **Overview of the tactile dataset setup.** In-domain data is collected using robotic exploration. Out-of-distribution data is collected using a different sensor unit, with human-operated exploration, introducing higher variance in contact force and angle.

variability, ordinal structure may remain more stable than absolute scaling. Figure 1 illustrates this motivation. Under a shift from Sensor A to Sensor B, pointwise predictions may undergo scale distortion, whereas pairwise learning focuses on preserving the relative ordering among roughness levels.

Nevertheless, most learning-based approaches to tactile material perception formulate the problem as either material classification [15]–[18] or absolute property regression [19]–[21]. Both formulations implicitly assume that tactile signals can be mapped to stable class boundaries or meaningful numerical targets. Moreover, models are typically evaluated under fixed sensor hardware and controlled exploration protocols, leaving the robustness of this absolute grounding under domain shift largely unexamined. While recent efforts highlight the importance of transferability and generalization in tactile perception [7], [22], the role of the learning objective itself in determining robustness has received comparatively little attention.

This raises a central question: *How does the choice of learning objective affect robustness in tactile roughness perception when absolute numerical grounding is unreliable?* Rather than proposing a new architecture, we isolate the effect of formulation by systematically comparing three common learning paradigms—classification, regression, and ranking-based learning [23]–[26]. Ranking-based objectives optimize relative ordering rather than absolute prediction error and are invariant to monotonic transformations of the output scale, making them conceptually well suited to settings where numerical grounding may drift.

Using a sandpaper dataset collected under controlled robotic exploration and evaluated under substantial domain shift—including changes in sensor hardware and exploration protocol—we perform a systematic empirical comparison across varying levels of training supervision. Our results show that while all paradigms achieve comparable performance under well-conditioned in-domain settings, their behavior diverges under data scarcity and domain shift. In particular, ranking-based learning more reliably preserves ordinal consistency when absolute numerical calibration becomes unstable, suggesting that aligning learning objectives with ordinal structure provides a principled path toward more robust tactile perception under realistic sensing variability.

II. Problem Formulation

A. *Label–Signal Mismatch in Tactile Roughness*

We formalize tactile roughness perception as a supervised learning problem in which a model maps a tactile observation x to a label y representing surface roughness. In regression-based formulations, this mapping implicitly assumes that there exists a stable function $f : \mathcal{X} \rightarrow \mathbb{R}$ such that predicted scalar outputs remain numerically aligned with ground-truth roughness values across sensing conditions. In this work, sensing conditions denote acquisition factors that shape the tactile observation, including the sensor unit and elastomer pad, applied normal force, contact angle, indentation depth, and exploration protocol.

Under realistic deployment scenarios, however, this assumption is fragile. Changes in sensing conditions—such as variations in contact dynamics, exploration strategy, or sensor-specific characteristics—can alter the distribution of tactile observations while leaving the underlying physical surface unchanged. As a result, the functional relationship between x and its associated scalar label y may shift across domains, even when y corresponds to a physically meaningful quantity. In other words, the mapping from tactile signal to absolute roughness value is condition-dependent.

This condition dependence creates a structural mismatch between supervision and signal. Pointwise regression objectives penalize deviations in absolute magnitude, presuming that identical surfaces should yield numerically consistent predictions regardless of sensing context. When this premise fails, minimizing absolute error does not necessarily preserve meaningful structure in the predictions.

Importantly, the instability induced by domain shift may not uniformly disrupt all relational information. While absolute magnitudes can drift across sensing contexts, the relative ordering between surfaces—i.e., whether surface i is rougher than surface j—may remain comparatively more stable. This observation suggests that, under domain variability, ordinal relationships impose weaker and potentially more robust constraints than absolute scalar targets.

One possible approach is to explicitly measure or control these factors and include them in the prediction model. However, this would require additional instrumentation, calibration, or tightly controlled exploration, which may not be available in practical tactile deployment. We therefore do not assume full observability of these nuisance variables, and instead ask whether the learning objective itself can provide robustness when such factors remain partially uncontrolled.

From this perspective, tactile roughness perception under sensing variability can be interpreted as a problem in which absolute numerical grounding is condition-sensitive, whereas ordinal structure may exhibit greater invariance. This motivates investigating learning objectives that prioritize relative ordering over absolute magnitude, forming the conceptual foundation of our comparative study.

B. Learning Objectives: Classification, Regression, and Ranking

By learning objective, we refer to the supervision format and loss function used to train the model, which determine whether the model is optimized for categorical agreement, absolute numerical accuracy, or relative ordering. We consider three learning paradigms commonly used in tactile perception. Let $x \in \mathcal{X}$ denote a tactile observation (e.g., a tactile image), and let $y \in \{1,\ldots,K\}$ denote its discrete roughness class, where K is the number of grit levels. For regression, we additionally define a scalar target $\tilde{y} \in \mathbb{R}$ obtained by normalizing the grit value to a fixed range. Depending on the learning paradigm, the model outputs either a probability distribution over classes or a scalar score.

- **Classification:**
  In classification, each tactile observation is assigned to a discrete roughness category. The model outputs class probabilities $p(x) = (p_1(x),\ldots,p_K(x))$ via a softmax layer, and training minimizes the standard cross-entropy loss:
  $$\mathcal{L}_{\text{cls}} = \mathbb{E}_{(x,y)}\left[-\log p_y(x)\right]. \tag{1}$$
  This formulation treats roughness labels as nominal categories and does not explicitly encode the ordered structure between grit levels.
- **Regression:**
  In regression, the model predicts a continuous scalar score $f(x) \in \mathbb{R}$ intended to approximate the normalized roughness target $\tilde{y}$. Training minimizes the mean squared error:
  $$\mathcal{L}_{\text{reg}} = \mathbb{E}_{(x,\tilde{y})}\left[\left(f(x)-\tilde{y}\right)^2\right]. \tag{2}$$
  This approach assumes that the numerical scale induced by grit values is meaningful and stable across sensing conditions. When this assumption is violated under domain shift, minimizing absolute prediction error may not preserve relative ordering.
- **Ranking:**
  In ranking-based learning, supervision is provided through ordered pairs of tactile observations. Let (x_i,x_j) denote a pair such that x_i corresponds to a rougher surface than x_j according to the ground-truth ordering. The model predicts scalar scores $f(x_i)$ and $f(x_j)$, and training minimizes a pairwise logistic ranking loss:
  $$\mathcal{L}_{\text{rank}} = \mathbb{E}_{(x_i,x_j)}\left[\log\left(1+\exp\left(-\left(f(x_i)-f(x_j)\right)\right)\right)\right]. \tag{3}$$
  Because the loss depends only on score differences, it focuses on relative ordering rather than absolute values.

TABLE I
IMPLEMENTATION DETAILS SHARED ACROSS ALL LEARNING PARADIGMS.

Backbone	ResNet-18 (ImageNet pretrained)
Input resolution	224×224 RGB
Normalization	ImageNet mean / std
Prediction head	Linear (task-specific)
Optimizer	Adam (lr = 1×10^{-4})
Batch size / Epochs	32 / 50
Train / Val split	80% / 20%
Train ratios	0.05, 0.25, 1.0
Random seeds	0, 1, 2
Ranking loss	Pairwise logistic (RankNet)
Score calibration	Linear fit on train split only
Hardware	NVIDIA GeForce RTX 4090
Software	PyTorch 2.0.1 (CUDA 11.7, cuDNN 8.5.0)

III. Method

A. Model Architecture

All learning paradigms share the same network architecture to isolate the effect of learning objectives. We use a ResNet-18 backbone pre-trained on ImageNet, with the final classification layer removed to obtain a 512-dimensional feature representation. A lightweight linear prediction head is attached to the backbone: a 5-way linear classifier for classification, a single-output linear regressor for regression, and a single-output linear scoring head for ranking. Aside from the prediction head, all architectural components are identical across methods.

B. Training and Implementation Details

Input tactile images are resized to 224×224 and normalized using ImageNet mean and standard deviation. The in-domain dataset consists of five sandpaper grit levels ({100, 180, 240, 320, 400}), with 1,919 images per class. Data is split into 80% training and 20% validation sets, and we study robustness under data scarcity via class-balanced subsampling at ratios of 0.05, 0.25, and 1.0.

All models are trained using Adam with a learning rate of 1×10^{-4} for 50 epochs and batch size 32. We fix random seeds for Python, NumPy, and PyTorch, and repeat all experiments over three seeds ({0,1,2}), yielding nine training configurations in total. All experiments are conducted on a single GPU (NVIDIA GeForce RTX 4090), using PyTorch 2.0.1 (CUDA 11.7, cuDNN 8.5.0), summarized in Table I.

For classification, we train with cross-entropy loss and select the checkpoint with the highest validation accuracy.

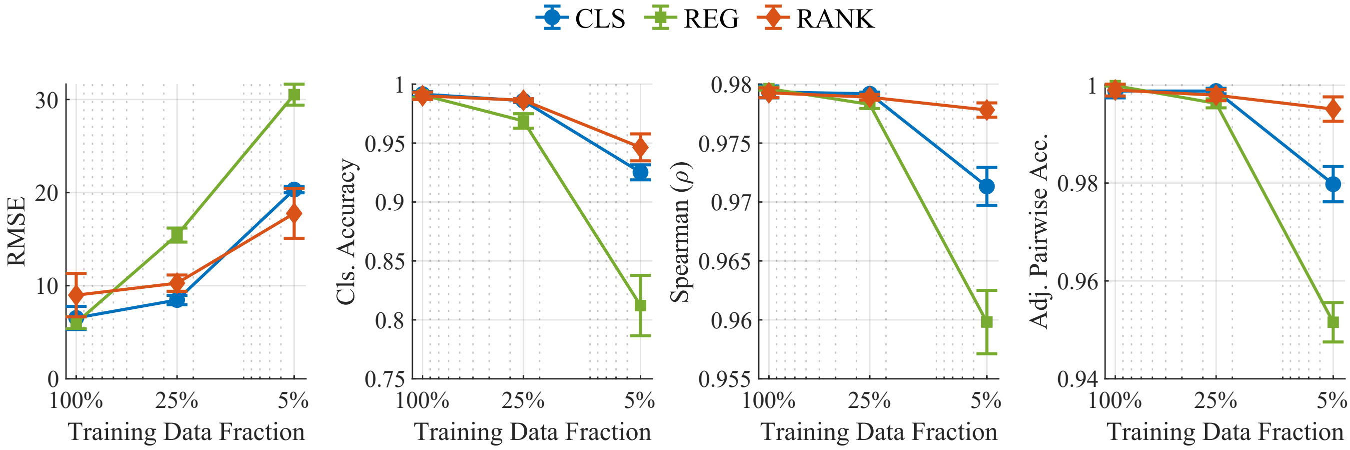

Fig. 3. **In-domain performance across training data fractions.** The left two plots show absolute metrics (RMSE and classification accuracy), and the right two plots show ordinal metrics (Spearman's ρ and adjacent pairwise accuracy). Performance is similar with sufficient data, with larger differences emerging at lower training fractions.

For regression, we minimize mean squared error on the normalized target $\tilde{y} \in [0,1]$ and select the checkpoint with the lowest validation RMSE. For ranking, we adopt a RankNet-style pairwise logistic loss on score differences. Training pairs are generated by randomly sampling pairs of images with different grit labels, with the number of sampled pairs proportional to the dataset size. We select the ranking checkpoint based on validation pairwise accuracy computed on randomly sampled validation pairs (up to 5,000 pairs), consistent with the RankNet training objective.

Since ranking models output scores on an arbitrary scale, we linearly map the predicted scores to grit values using the in-domain training split. This calibration is applied unchanged to validation and OOD data. Absolute metrics (RMSE/MAE) are computed on the calibrated outputs, whereas ordinal metrics (Spearman and pairwise accuracy) depend only on relative ordering and therefore do not require calibration.

IV. Evaluation

A. In-Domain and Out-of-Distribution Datasets

We evaluate tactile roughness perception under both in-domain and out-of-distribution (OOD) conditions. Figure 2 provides an overview of the data collection setup and illustrates the key differences between in-domain and OOD conditions. In both settings, the objects are commercially available foam-backed abrasive blocks (siasponge, sia Abrasives) with five grit levels (100, 180, 240, 320, and 400), forming a discrete ordinal roughness scale.

The in-domain dataset is collected via robot-controlled exploration using a 7 DoF robot arm (Gen2, Kinova inc.), and attached vision-based tactile sensor (DIGIT, GelSight). Data is acquired through repeated indentations over multiple contact locations and normal force levels, resulting in a relatively controlled sensing and interaction regime. The total $N = 9{,}595$ images, which is 1,919 images for each of the five grit labels, are collected.

To assess robustness under realistic sensing variability, we additionally construct an OOD evaluation set collected under substantially different conditions: a different unit of the same DIGIT tactile sensor, a different elastomer pad interface, human-operated rather than robot-controlled exploration, and uncontrolled variations in contact force, contact angle, and indentation depth. Compared to the in-domain setting, these factors introduce simultaneous shifts in both sensor response and exploration dynamics. The OOD evaluation set contains a total of $N = 1{,}500$ images, with 300 images per class.

B. Robustness Evaluation Protocol

We evaluate robustness by systematically stressing two factors: (i) limited training data and (ii) domain shift at evaluation time. Concretely, we train models at three class-balanced subsampling ratios (0.05, 0.25, and 1.0) and repeat experiments over three random seeds (0, 1, 2), yielding nine training configurations.

The in-domain dataset is partitioned into 80% training and 20% validation splits. Subsampling is applied only to the training split, while the validation split remains fixed across all experiments. The OOD dataset is used strictly for evaluation and is never accessed during training or calibration.

Each trained model is evaluated on both the in-domain validation split and the OOD set, enabling direct comparison of generalization gaps across objectives.

For ranking-based learning, supervision is provided through pairwise comparisons. To avoid information leakage, all calibration used for mapping ranking scores to grit values is fit on the in-domain training split only and then applied unchanged to validation and OOD data. We treat this as a critical protocol choice because calibration can otherwise artificially reduce OOD error by exploiting evaluation labels.

C. Evaluation Metrics

We use complementary metrics to separately evaluate absolute numerical fidelity and ordinal consistency. This

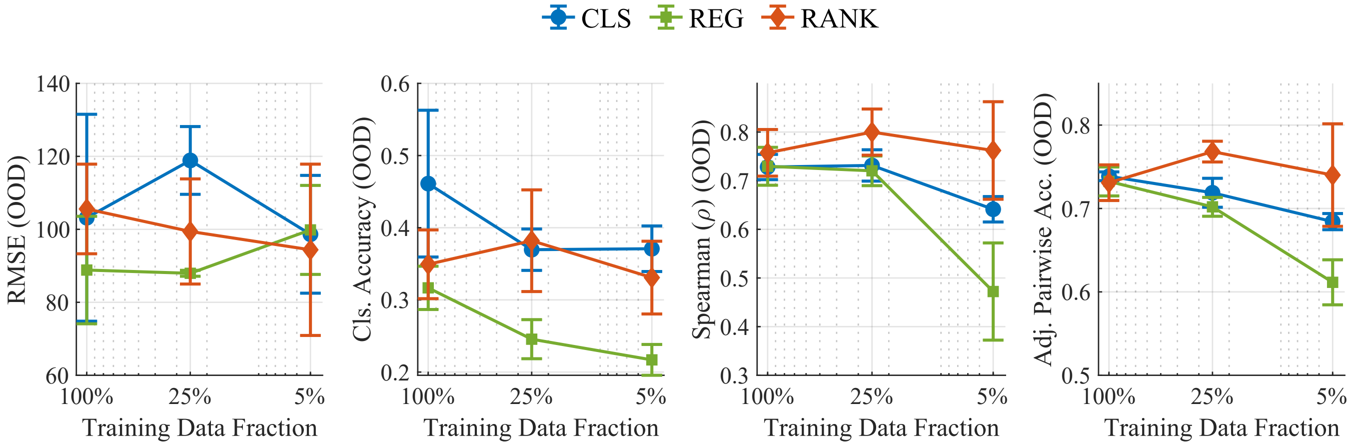

Fig. 4. **Out-of-distribution performance across training fractions.** The plots show RMSE, classification accuracy, Spearman's ρ, and adjacent pairwise accuracy for classification (CLS), regression (REG), and ranking (RANK). Ranking-based learning achieves stronger ordinal consistency under domain shift, especially in low-data regimes. Error bars denote standard deviation over three seeds.

distinction is central to our study: pointwise objectives can score well on absolute error while failing to preserve ordering under scale drift.

- **Absolute Metrics:**
  We report root mean squared error (RMSE) and classification accuracy between model predictions and ground-truth roughness labels. RMSE measures numerical deviation on the scalar roughness scale, and classification accuracy evaluates discrete label agreement. These metrics quantify fidelity to absolute target values under both continuous and categorical formulations.
- **Ordinal Metrics:** To evaluate consistency in relative ordering, we report Spearman's rank correlation and pairwise accuracy. Spearman's rank correlation measures agreement between the ranking induced by predicted scores and the ranking induced by ground-truth labels. Given N evaluation samples with predicted scores $\{f(x_n)\}_{n=1}^{N}$ and roughness labels $\{y_n\}_{n=1}^{N}$, Spearman's ρ is defined as

$$\rho = \operatorname{corr}(\operatorname{rank}(f(x)), \operatorname{rank}(y)), \tag{4}$$

  where $\operatorname{rank}(\cdot)$ assigns average ranks to ties and $\operatorname{corr}(\cdot,\cdot)$ denotes Pearson correlation. Because ρ is invariant to monotonic transformations of scores, it directly reflects ordinal consistency rather than calibration.
  In addition, we report pairwise accuracy, defined as the fraction of sample pairs whose predicted ordering matches the ground-truth ordering. We compute (i) all-pairs accuracy over all valid grit pairs and (ii) adjacent pairwise accuracy considering only neighboring grit levels. The adjacent metric emphasizes fine-grained discrimination and avoids dominance by trivially separable pairs, making it a more sensitive probe of local ordinal robustness.

V. Results

A. In-Domain Evaluation

In-domain performance across training fractions is shown in Figure 3. We compare classification (CLS), regression (REG), and ranking (RANK) using four metrics: RMSE and classification accuracy (absolute), and Spearman's ρ and adjacent pairwise accuracy (ordinal). All values are reported as the mean over three random seeds.

- **Train ratio = 1.0.**
  RMSE (CLS=6.54, REG=6.02, RANK=8.98). Classification accuracy (CLS=0.991, REG=0.991, RANK=0.990). Spearman (CLS=0.979, REG=0.980, RANK=0.979). Adjacent pairwise accuracy (CLS=0.9988, REG=0.9999, RANK=0.9990).
  All three methods perform similarly under full supervision, with only minor differences across both absolute and ordinal metrics.
- **Train ratio = 0.25.**
  RMSE (CLS=8.47, REG=15.44, RANK=10.27). Classification accuracy (CLS=0.986, REG=0.969, RANK=0.986). Spearman (CLS=0.979, REG=0.978, RANK=0.979). Adjacent pairwise accuracy (CLS=0.9988, REG=0.9963, RANK=0.9980).
  Compared to full supervision, regression exhibits a substantial increase in RMSE and a noticeable drop in classification accuracy, while classification and ranking remain relatively stable. Ordinal metrics remain high across all models.
- **Train ratio = 0.05.**
  RMSE (CLS=20.32, REG=30.53, RANK=17.76). Classification accuracy (CLS=0.925, REG=0.812, RANK=0.946). Spearman (CLS=0.971, REG=0.960, RANK=0.978). Adjacent pairwise accuracy (CLS=0.980, REG=0.952, RANK=0.995).
  Under limited supervision, performance gaps widen. Ranking achieves the lowest RMSE and highest classification accuracy. Ordinal consistency also shows clearer

separation, with ranking maintaining higher Spearman and adjacent pairwise accuracy than both classification and regression.

Overall, in-domain differences between objectives are small under sufficient data, but become more visible as supervision decreases. Notably, the relative ordering of models varies across training fractions, particularly for RMSE, indicating that objective choice interacts with data availability even without domain shift.

B. Out-of-Distribution Evaluation

OOD ordinal performance across training fractions is summarized in Figure 4. All four metrics are reported for completeness, and Table II provides a compact comparison at train ratio = 0.25.

- **Train ratio = 1.0.** RMSE (CLS=103.18, REG=88.84, RANK=105.58). Classification accuracy (CLS=0.461, REG=0.317, RANK=0.349). Spearman (CLS=0.728, REG=0.730, RANK=0.757). Adjacent pairwise accuracy (CLS=0.739, REG=0.732, RANK=0.731).
  Under domain shift with full supervision, regression attains the lowest RMSE, while ranking achieves the highest Spearman. Differences in adjacent pairwise accuracy are comparatively small.
- **Train ratio = 0.25.** RMSE (CLS=118.88, REG=87.95, RANK=99.42). Classification accuracy (CLS=0.370, REG=0.246, RANK=0.382). Spearman (CLS=0.731, REG=0.720, RANK=0.800). Adjacent pairwise accuracy (CLS=0.719, REG=0.702, RANK=0.768).
  At this regime, regression minimizes RMSE, while ranking attains the highest classification accuracy and strongest ordinal consistency. The separation in Spearman and adjacent pairwise accuracy becomes more pronounced.
- **Train ratio = 0.05.** RMSE (CLS=98.64, REG=99.84, RANK=94.39). Classification accuracy (CLS=0.371, REG=0.217, RANK=0.331). Spearman (CLS=0.641, REG=0.472, RANK=0.762). Adjacent pairwise accuracy (CLS=0.684, REG=0.611, RANK=0.740).
  With both domain shift and limited supervision, ranking achieves the lowest RMSE and highest ordinal metrics. Regression exhibits the largest degradation in Spearman relative to its in-domain performance.

Across OOD settings, lower RMSE does not consistently correspond to higher ordinal consistency. Figure 5 illustrates this trade-off between absolute calibration (RMSE) and ordinal structure (Spearman), highlighting that objectives directly optimizing relative ordering retain stronger ordinal behavior under domain shift.

VI. Discussion and Conclusion

This study isolates how the choice of learning objective influences robustness in tactile roughness perception when absolute numerical grounding becomes unstable. By keeping the network architecture and training protocol fixed, we attribute differences in generalization behavior primarily to the supervision formulation: pointwise regression, categorical classification, or pairwise ranking.

A possible explanation is that domain shift changes not only the distribution of tactile images but also the effective scale relating tactile responses to numerical roughness labels. Ranking-based learning relaxes the need for this scale to remain globally stable, because it optimizes pairwise relations rather than pointwise numerical agreement.

Across both limited supervision and domain shift, a consistent pattern emerges. Regression-based learning remains competitive in minimizing absolute numerical error, yet its ordinal structure degrades substantially under out-of-distribution (OOD) shift. In the most challenging 5% training regime, regression's Spearman correlation drops from $\rho = 0.960$ (in-domain) to $\rho = 0.472$ (OOD), reflecting a severe loss of ordinal consistency. In contrast, ranking-based learning decreases from $\rho = 0.978$ to $\rho = 0.762$, demonstrating a substantially smaller degradation. This more gradual decline indicates that the structure preserved by ranking objectives remains more stable under sensing variability.

This behavior can be understood from the properties of the learning objectives themselves. Regression minimizes pointwise deviations between predicted scores and scalar targets. When domain shift induces a condition-dependent distortion of the mapping between tactile signals and numerical labels—such as an unknown affine or nonlinear but monotonic transformation—the regression objective penalizes even order-preserving transformations. In contrast, the pairwise logistic ranking loss depends only on score differences and is invariant to monotonic transformations of the output scale. As long as the relative ordering between samples is preserved, the ranking objective remains aligned with the supervision signal. Thus, when absolute calibration

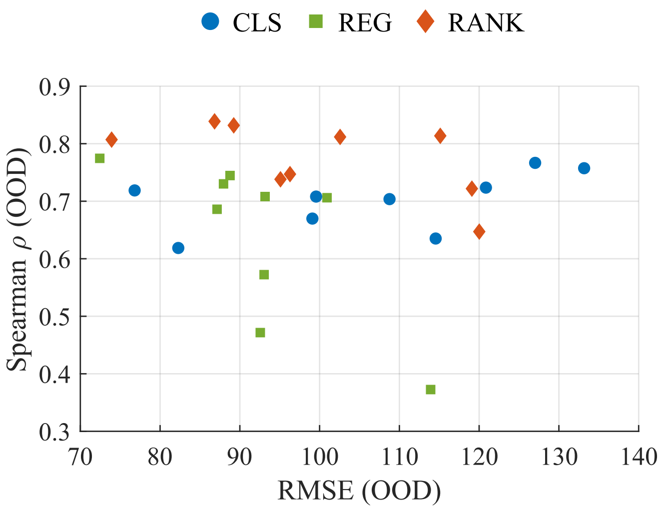

Fig. 5. **Trade-off between absolute accuracy (RMSE) and ordinal consistency (Spearman ρ) under domain shift.** Each point corresponds to a model trained with a specific random seed and training data fraction. Lower RMSE does not necessarily imply better ordinal consistency.

TABLE II
QUANTITATIVE COMPARISON ON OUT-OF-DISTRIBUTION (OOD) DATA AT TRAIN RATIO = 0.25 (MEAN ± STD OVER 3 SEEDS).

Method	RMSE ↓	Cls. Acc. ↑	Spearman ↑	Adj. Pairwise ↑
Classification	118.88 ± 7.58	0.370 ± 0.023	0.731 ± 0.026	0.719 ± 0.014
Regression	**87.95** ± 0.66	0.246 ± 0.022	0.720 ± 0.025	0.702 ± 0.009
Ranking	99.42 ± 11.76	**0.382** ± 0.058	**0.800** ± 0.039	**0.768** ± 0.010

drifts but ordinal relationships remain comparatively stable, ranking imposes structurally weaker and more robust constraints.

Although pairwise accuracy is naturally aligned with ranking-based training, the same trend is observed in Spearman's rank correlation, which is objective-agnostic and depends only on induced ordering. The consistent advantage of ranking in both ordinal metrics suggests that the observed robustness is not merely an artifact of metric alignment, but reflects genuine preservation of relational structure under domain shift.

Importantly, we do not claim that ordinal objectives universally replace regression or classification. When absolute physical estimation is reliable and numerically meaningful across sensing conditions, pointwise regression remains appropriate. However, our results indicate that when scalar labeling schemes are used in settings where the mapping between sensory signals and numerical targets becomes condition-dependent, directly optimizing ordinal structure provides a more stable inductive bias. In such cases, robustness depends less on the semantic interpretation of the scalar label itself and more on how the learning objective interacts with scale variability.

The preserved ordinal consistency of ranking-based models may also offer practical advantages for deployment. A model that maintains stable relative ordering under domain shift could be used in a two-stage strategy: first learn a representation that preserves ordinal structure across sensing conditions, and then re-ground the output scale in a new domain using a small number of reference surfaces. While we do not explicitly evaluate such re-grounding strategies here, the observed stability suggests that ranking-based representations may serve as a robust anchor for lightweight post-deployment calibration.

This work has several limitations. First, our experiments are conducted on a controlled sandpaper roughness dataset; broader material properties, more complex textures, and other tactile attributes such as hardness, compliance, and friction should be examined in future work. Second, we evaluate perception in isolation rather than within closed-loop manipulation tasks such as grasp stabilization or slip control. Understanding how ordinally trained representations influence downstream control remains an important open question. Finally, our OOD setting intentionally combines several practical sources of shift, including sensor-unit and pad differences and exploration variability. While this provides a realistic stress test, it does not disentangle which factor contributes most strongly to the degradation of each objective. Controlled single-factor shift studies would provide finer-grained insight into the sources of scale instability.

In conclusion, when tactile perception operates under realistic sensing variability—where absolute numerical grounding may drift but relational structure remains meaningful—aligning learning objectives with ordinal relationships provides a more robust foundation than optimizing absolute numerical error alone. Rather than treating roughness prediction purely as a scalar estimation problem, our results suggest that embracing its inherently comparative structure can yield more stable generalization under domain shift.

REFERENCES

[1] W. Yuan, S. Dong, and E. H. Adelson, "Gelsight: High-resolution robot tactile sensors for estimating geometry and force," *Sensors*, vol. 17, no. 12, p. 2762, 2017.

[2] E. Donlon, S. Dong, M. Liu, J. Li, E. Adelson, and A. Rodriguez, "Gelslim: A high-resolution, compact, robust, and calibrated tactile-sensing finger," in *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2018, pp. 1927–1934.

[3] M. Lambeta, P.-W. Chou, S. Tian, B. Yang, B. Maloon, V. R. Most, D. Stroud, R. Santos, A. Byagowi, G. Kammerer *et al.*, "Digit: A novel design for a low-cost compact high-resolution tactile sensor with application to in-hand manipulation," *IEEE Robotics and Automation Letters*, vol. 5, no. 3, pp. 3838–3845, 2020.

[4] B. Ward-Cherrier, N. Pestell, L. Cramphorn, B. Winstone, M. E. Giannaccini, J. Rossiter, and N. F. Lepora, "The tactip family: Soft optical tactile sensors with 3d-printed biomimetic morphologies," *Soft robotics*, vol. 5, no. 2, pp. 216–227, 2018.

[5] A. Alspach, K. Hashimoto, N. Kuppuswamy, and R. Tedrake, "Soft-bubble: A highly compliant dense geometry tactile sensor for robot manipulation," in *2019 2nd IEEE International Conference on Soft Robotics (RoboSoft)*. IEEE, 2019, pp. 597–604.

[6] S. Luo, J. Bimbo, R. Dahiya, and H. Liu, "Robotic tactile perception of object properties: A review," *Mechatronics*, vol. 48, pp. 54–67, 2017.

[7] S. Luo, N. F. Lepora, W. Yuan, K. Althoefer, G. Cheng, and R. Dahiya, "Tactile robotics: An outlook," *IEEE Transactions on Robotics*, 2025.

[8] H. Li, Y. Lin, C. Lu, M. Yang, E. Psomopoulou, and N. F. Lepora, "Classification of vision-based tactile sensors: A review," *IEEE Sensors Journal*, 2025.

[9] B. Fang, J. Zhao, N. Liu, Y. Sun, S. Zhang, F. Sun, J. Shan, and Y. Yang, "Force measurement technology of vision-based tactile sensor," *Advanced Intelligent Systems*, vol. 7, no. 1, p. 2400290, 2025.

[10] S. Stevens and J. R. Harris, "The scaling of subjective roughness and smoothness." *Journal of experimental psychology*, vol. 64, no. 5, p. 489, 1962.

[11] M. Kahrimanovic, W. M. Bergmann Tiest, and A. M. Kappers, "Context effects in haptic perception of roughness," *Experimental brain research*, vol. 194, no. 2, pp. 287–297, 2009.

[12] R. D. Roberts, A. R. Loomes, H. A. Allen, M. Di Luca, and A. M. Wing, "Contact forces in roughness discrimination," *Scientific reports*, vol. 10, no. 1, p. 5108, 2020.

[13] G. Ekman, J. Hosman, and B. Lindstrom, "Roughness, smoothness, and preference: a study of quantitative relations in individual subjects." *Journal of Experimental Psychology*, vol. 70, no. 1, p. 18, 1965.

[14] R. D. Roberts, "Roughness perception across the hands," *Attention, Perception, & Psychophysics*, vol. 75, no. 6, pp. 1306–1317, 2013.

[15] S. S. Baishya and B. Bäuml, "Robust material classification with a tactile skin using deep learning," in *2016 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2016, pp. 8–15.

[16] Y. Gao, L. A. Hendricks, K. J. Kuchenbecker, and T. Darrell, “Deep learning for tactile understanding from visual and haptic data,” in *2016 IEEE international conference on robotics and automation (ICRA)*. IEEE, 2016, pp. 536–543.

[17] S. Luo, W. Yuan, E. Adelson, A. G. Cohn, and R. Fuentes, “Vitac: Feature sharing between vision and tactile sensing for cloth texture recognition,” in *2018 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2018, pp. 2722–2727.

[18] F. Pastor, J. M. Gandarias, A. J. García-Cerezo, and J. M. Gómez-de Gabriel, “Using 3d convolutional neural networks for tactile object recognition with robotic palpation,” *Sensors*, vol. 19, no. 24, p. 5356, 2019.

[19] W. Yuan, C. Zhu, A. Owens, M. A. Srinivasan, and E. H. Adelson, “Shape-independent hardness estimation using deep learning and a gelsight tactile sensor,” in *2017 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 2017, pp. 951–958.

[20] Y. Yan, Z. Hu, Y. Shen, and J. Pan, “Surface texture recognition by deep learning-enhanced tactile sensing,” *Advanced Intelligent Systems*, vol. 4, no. 1, p. 2100076, 2022.

[21] Y. Song, S. Lv, F. Wang, and M. Li, “Hardness-and-type recognition of different objects based on a novel porous graphene flexible tactile sensor array,” *Micromachines*, vol. 14, no. 1, p. 217, 2023.

[22] Z. Chen, N. Ou, X. Zhang, Z. Wu, Y. Zhao, Y. Wang, N. Lepora, L. Jamone, J. Deng, and S. Luo, “General force sensation for tactile robot,” *arXiv preprint arXiv:2503.01058*, 2025.

[23] R. Herbrich, T. Graepel, and K. Obermayer, “Support vector learning for ordinal regression,” in *1999 Ninth International Conference on Artificial Neural Networks ICANN 99.(Conf. Publ. No. 470)*, vol. 1. IET, 1999, pp. 97–102.

[24] C. Burges, T. Shaked, E. Renshaw, A. Lazier, M. Deeds, N. Hamilton, and G. Hullender, “Learning to rank using gradient descent,” in *Proceedings of the 22nd international conference on Machine learning*, 2005, pp. 89–96.

[25] D. Parikh and K. Grauman, “Relative attributes,” in *2011 International conference on computer vision*. IEEE, 2011, pp. 503–510.

[26] L. Xu, J. Li, W. Lin, Y. Zhang, Y. Zhang, and Y. Yan, “Pairwise comparison and rank learning for image quality assessment,” *Displays*, vol. 44, pp. 21–26, 2016.